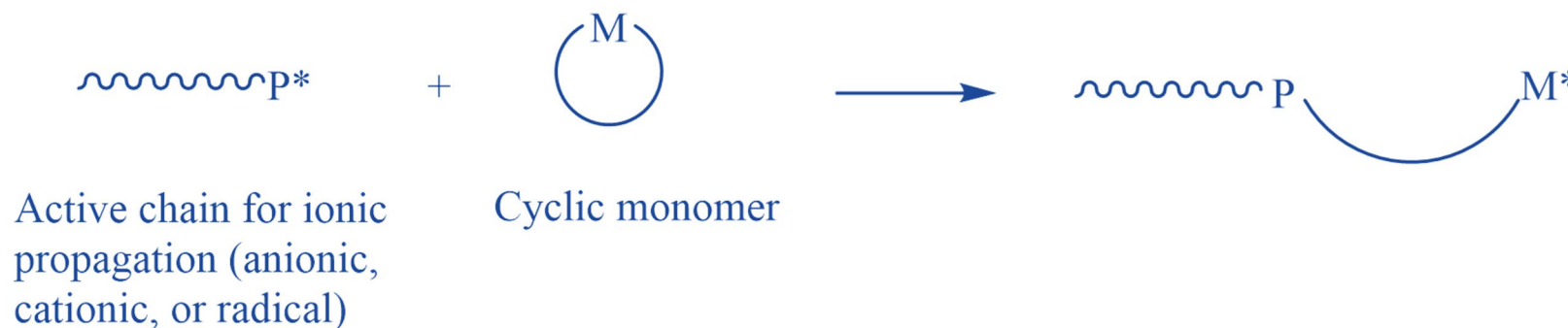


CHAPTER 4

Ring-opening polymerizations

Main Concept

ROPs rely on a chain-growth mechanism, in which monomers are cyclic and will be opened once incorporated into the growing chain.



Important features

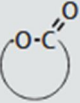
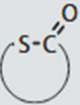
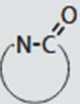
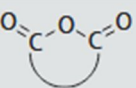
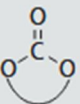
- Chain polymerization → high-molar-mass polymers, possibility to control dispersity and functionality
- Opened ring goes into polymer backbone → access to degradable polymers

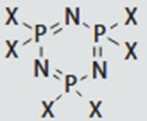
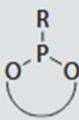
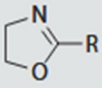
Thermodynamic driving force

- Enthalpy = relief of ring strain (usually up to ca. 8 atoms)
- Entropy = increase of conformational freedom (from ca. 14 atoms on)

Ring-Opening Polymerizations

Monomer Examples

Name	Structure	Ring size	Initiator type
Olefin		4, 5, 8	Metathesis
Ether		3, 4, 5, 7	Cationic, anionic
Thioether		3, 4	Cationic, anionic
Amine		3, 4, 7	Cationic
Lactone		4, 6, 7, 8	Cationic, anionic
Thiolactone		4-8	Cationic, anionic
Lactam		≥ 4	Cationic, anionic
Disulfide		≥ 8	Radical
Anhydride		5 and ≥ 7	Anionic
Carbonate		6, 7, 8 and ≥ 20	Anionic

Name	Structure	Ring size	Initiator type
Silicone		6, 8 and ≥ 10	Cationic, anionic
Phosphazene		6	Cationic
Phosponite		3, 5, 6, 7	Anionic
Oxazoline		5	Cationic

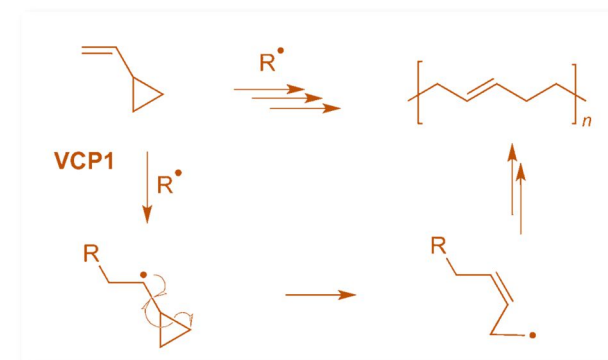
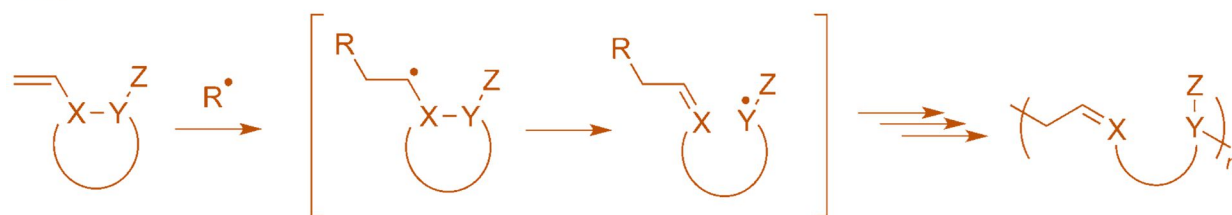
Polymer Chemistry; Ed. S. Koltzenburg, M. Maskos, O. Nuyken, Springer

Ring-Opening Polymerizations

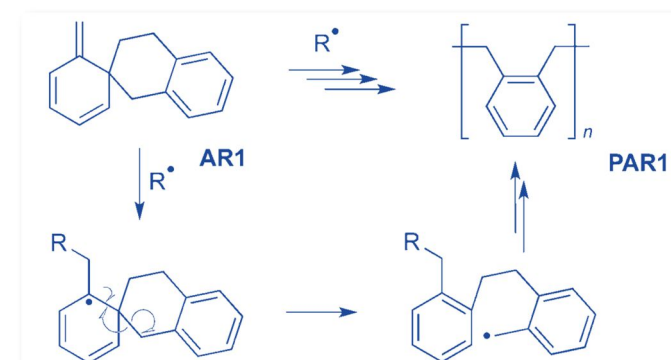
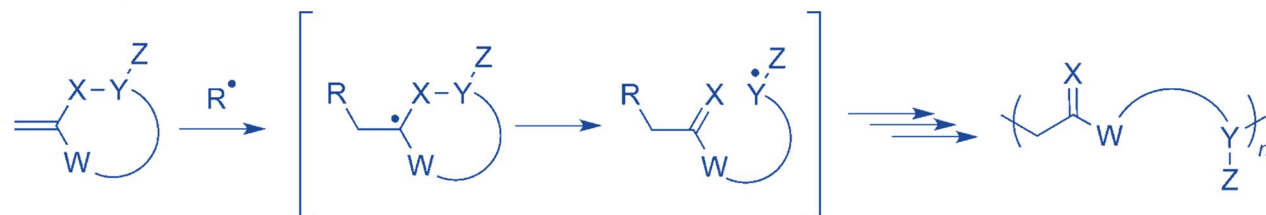
Radical Ring-Opening Polymerization (rROP)

Radical addition, generally on a double bond, followed by ring opening and formation of a new radical

Vinyl



Exo-Methylene

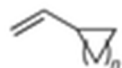


Ring-Opening Polymerizations

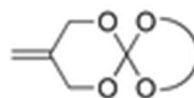
Radical Ring-Opening Polymerization (rROP)

Radical addition, generally on a double bond, followed by ring opening and formation of a new radical

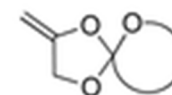
vinyl cycloalkane (VCA)



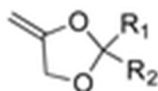
spiro-ortho-carbonate (SOC)



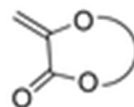
spiro-ortho-ester (SOE)



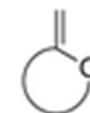
cyclic vinyl acetal (CVA)



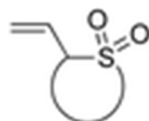
cyclic α -oxyacrylate (C α OA)



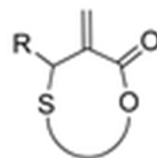
cyclic vinyl ether (CVE)



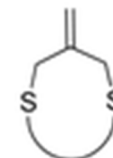
cyclic vinyl sulfone (CVS)



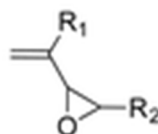
sulfide cyclic methacrylate (SCM)



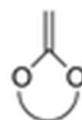
cyclic allylic sulfide (CAS)



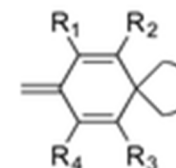
vinyl oxirane (VO)



cyclic ketene acetal (CKA)



cyclic monomer with ring-opening driven by aromatization (AR)



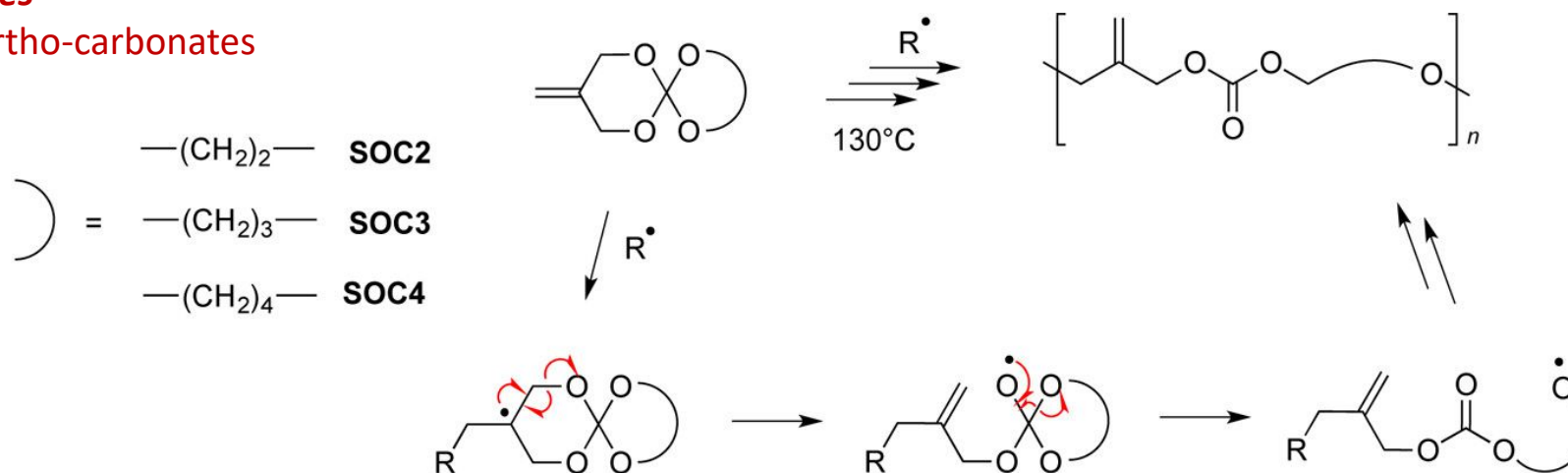
Ring-Opening Polymerizations

Radical Ring-Opening Polymerization (rROP)

Radical addition, generally on a double bond, followed by ring opening and formation of a new radical

Examples

spiro-ortho-carbonates



	monomer	polymerization conditions	double ring-opening (% r)	refs
SOC1		130 °C, sol.1.25M, 20h	5% , + cross-linking	37
SOC5		130 °C, bulk, 20h 180 °C, bulk, 20h	4% 24%	38
SOC6		130 °C, 20h 180 °C, bulk, 20h	8% bulk, 9% sol. 55%	38
SOC7		130 °C, bulk, 20h 180 °C, bulk, 20h	34% 86%	38

Chem. Rev. 2017, 117, 1319

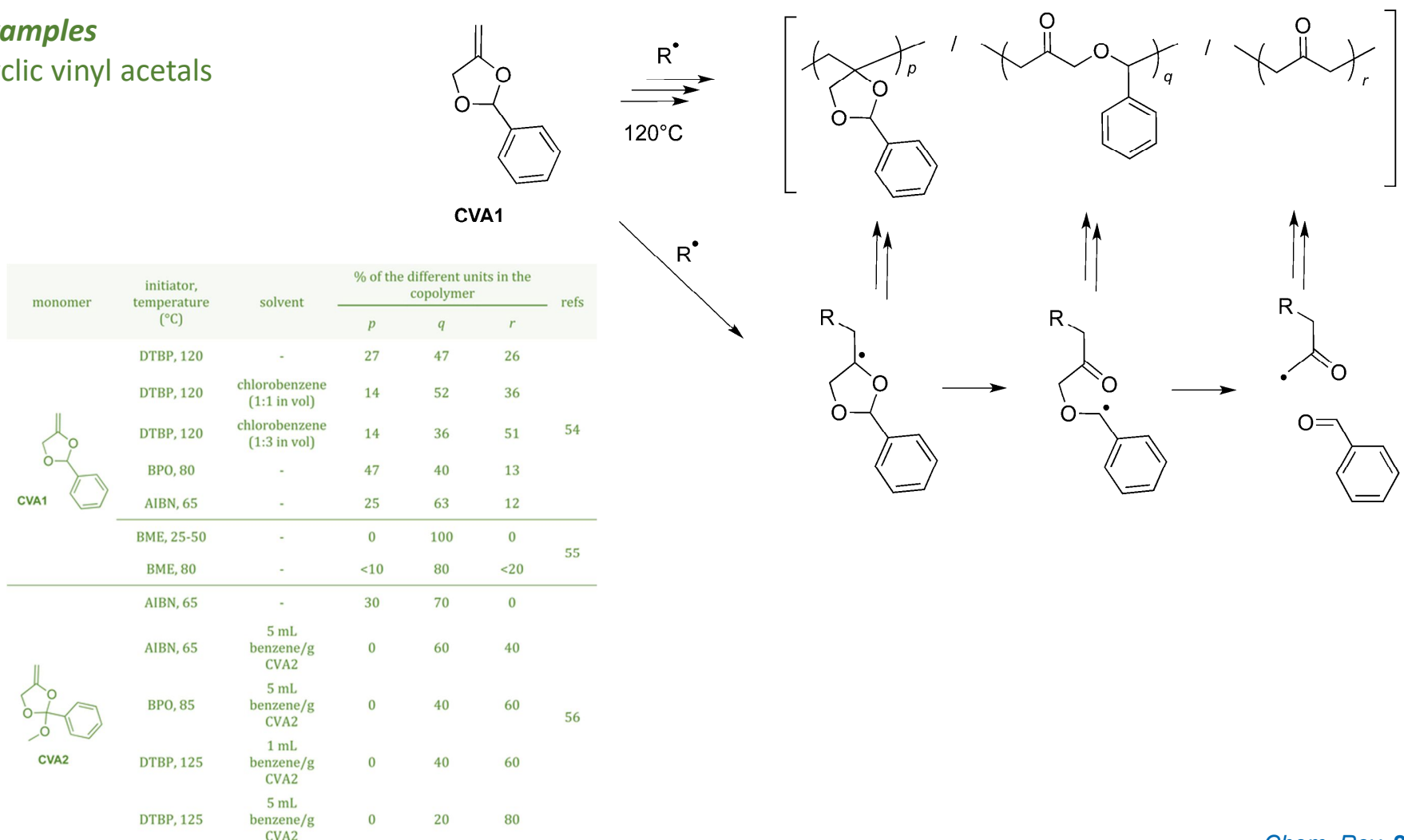
Ring-Opening Polymerizations

Radical Ring-Opening Polymerization (rROP)

Radical addition, generally on a double bond, followed by ring opening and formation of a new radical

Examples

Cyclic vinyl acetals



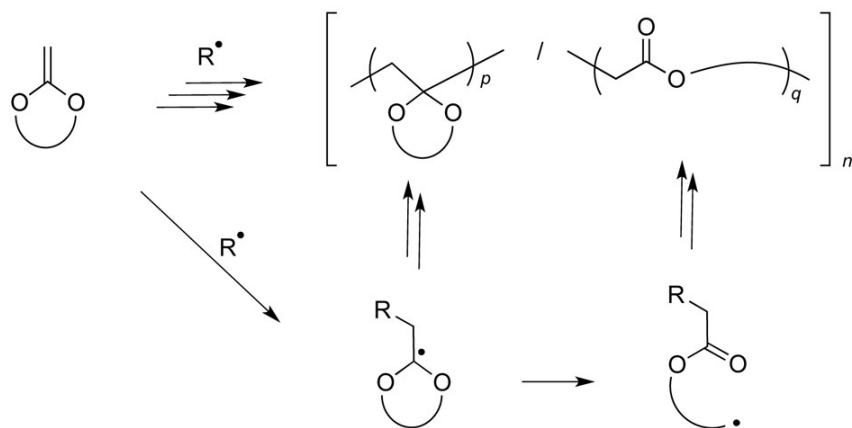
Ring-Opening Polymerizations

Radical Ring-Opening Polymerization (rROP)

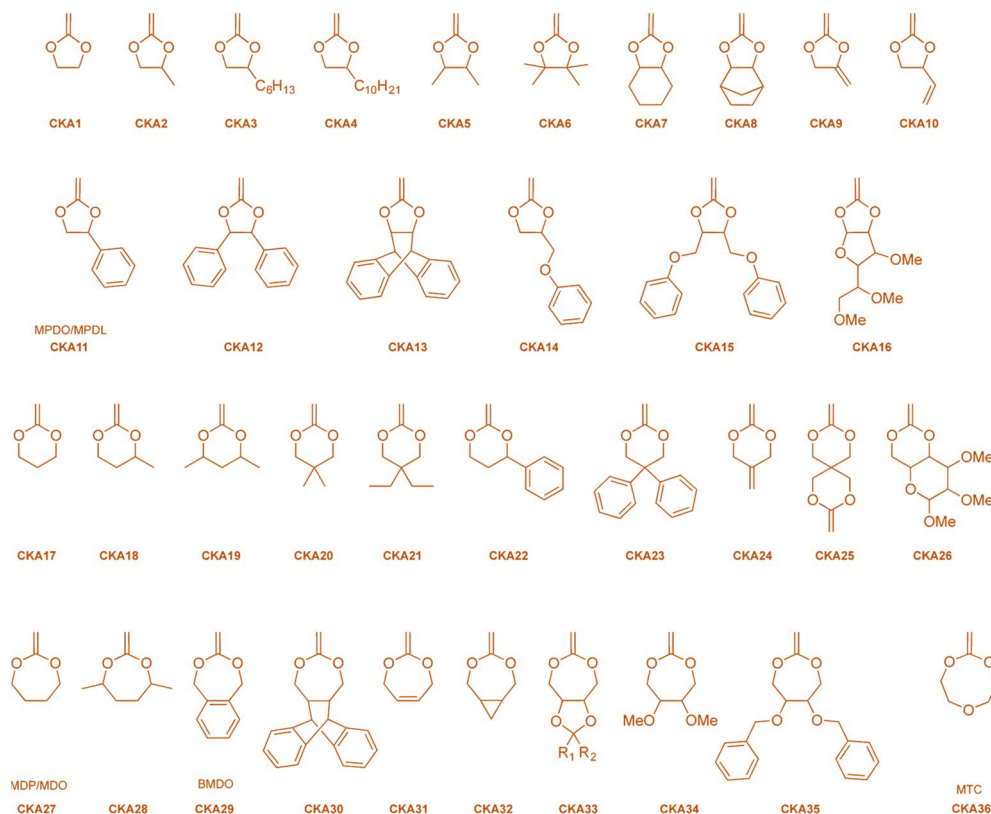
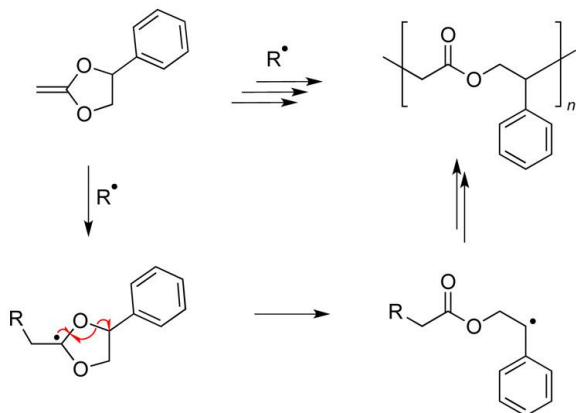
Radical addition, generally on a double bond, followed by ring opening and formation of a new radical

Examples

Cyclic ketene acetals



e.g., 2-methylene-4-phenyl-1,3-dioxolane



Chem. Rev. 2017, 117, 1319

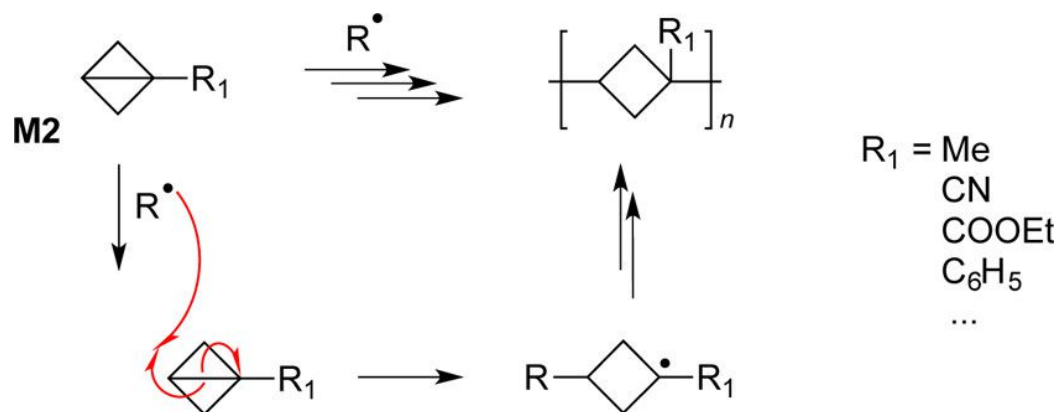
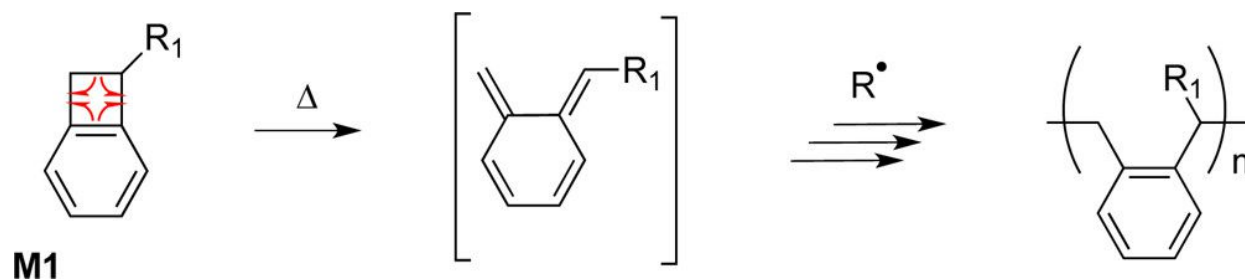
Ring-Opening Polymerizations

Radical Ring-Opening Polymerization (rROP)

Radical addition, generally on a double bond, followed by ring opening and formation of a new radical

Examples

Monomers without C=C double bond



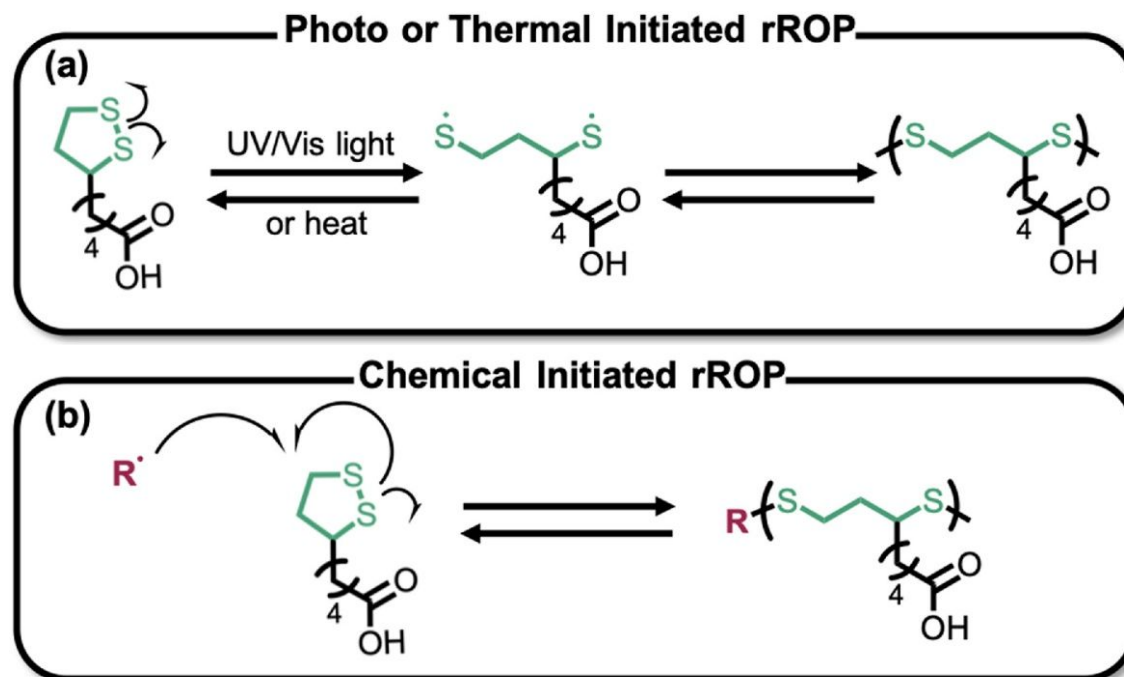
Ring-Opening Polymerizations

Radical Ring-Opening Polymerization (rROP)

Radical addition, generally on a double bond, followed by ring opening and formation of a new radical

Examples

Monomers without C=C double bond



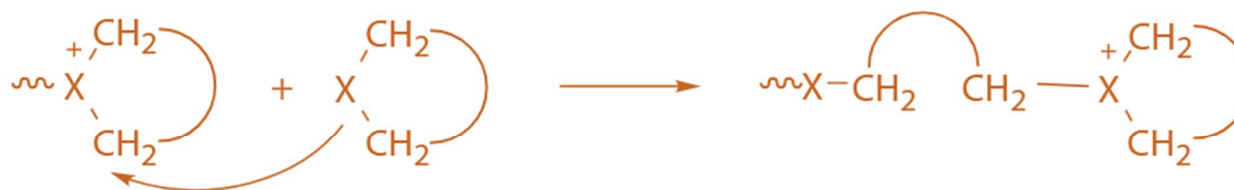
Ring-Opening Polymerizations

Cationic Ring-Opening Polymerization (cROP)

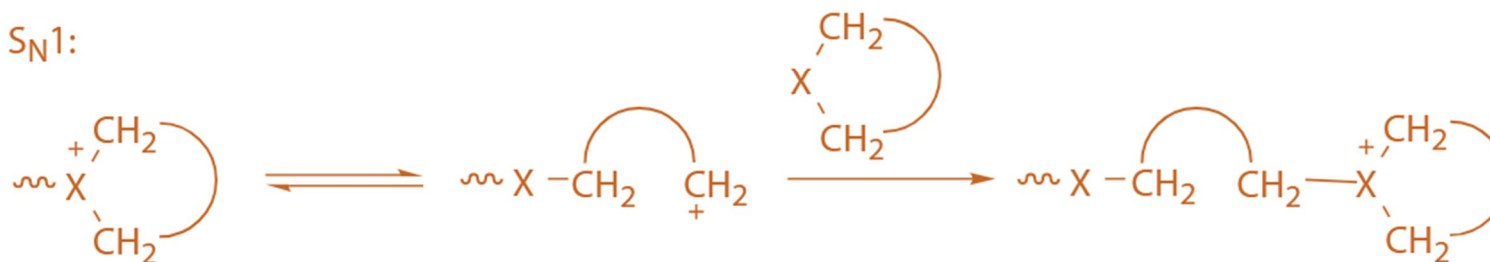
Two general mechanisms can be used:

(i) S_N -type mechanism

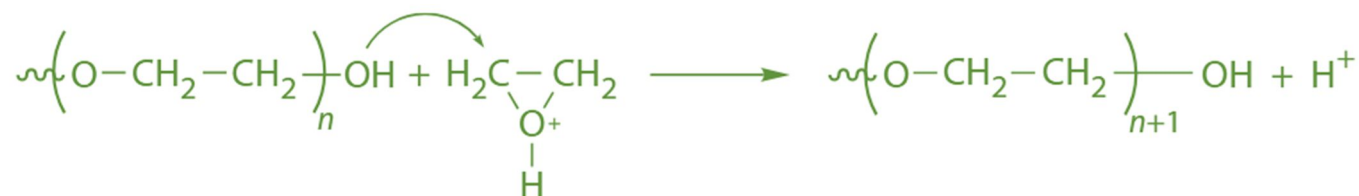
S_N2 :



S_N1 :



(ii) activated-monomer mechanism



Cationic Ring-Opening Polymerization (cROP)

Initiation

In principle, similar initiators as for the olefin cationic polymerization can be used.

- **Brønsted acids** (HCl , H_2SO_4 , HClO_4 , HOSO_2CF_3)
- **carbenium ions**
- **onium ions**
- **covalent initiators**
- **Lewis acids (+ co-initiator)**
- **photoinitiators**

Some cROP variants are living but there are often inherent side-reactions.

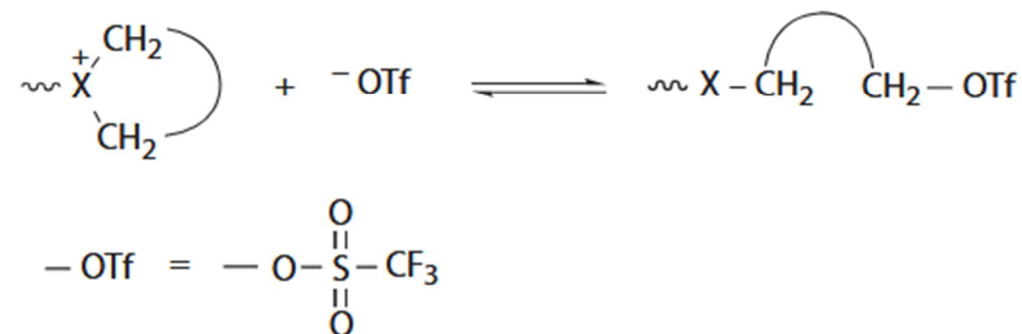
1. Backbiting



2. Separation of rings



3. Collapse of counter ion



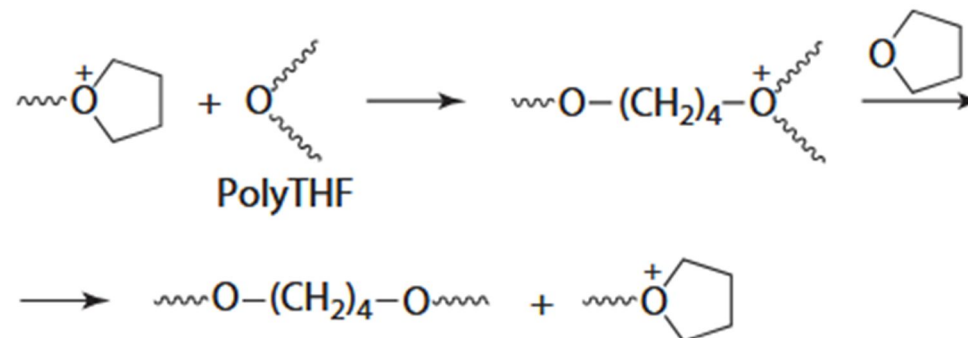
Ring-Opening Polymerizations

Cationic Ring-Opening Polymerization (cROP)

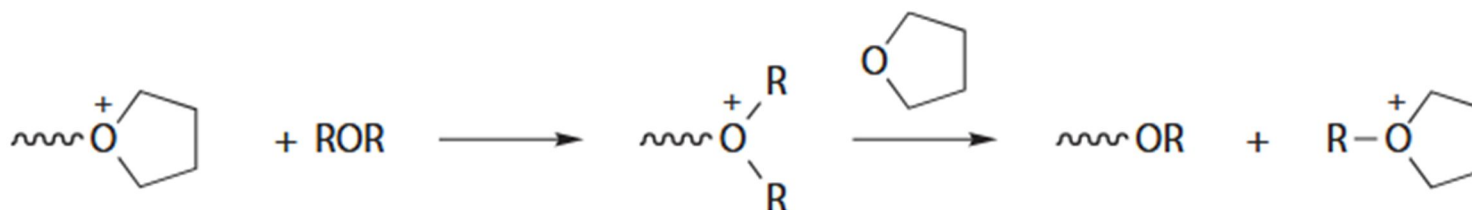
Some cROP variants are living but there are often inherent side-reactions.

At high monomer conversions, i.e., $[M]$ low, these reactions lead to termination, rather than transfer.

Termination by alkylation



this can be done voluntarily... addition of a low-molar-mass dialkyl ether



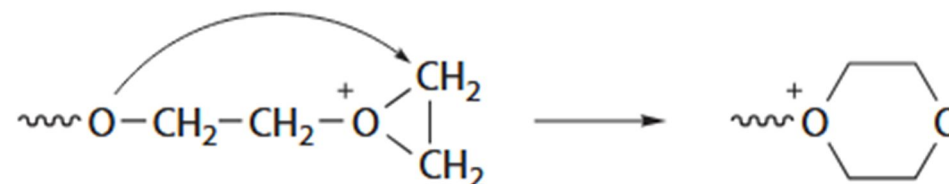
Ring-Opening Polymerizations

Cationic Ring-Opening Polymerization (cROP)

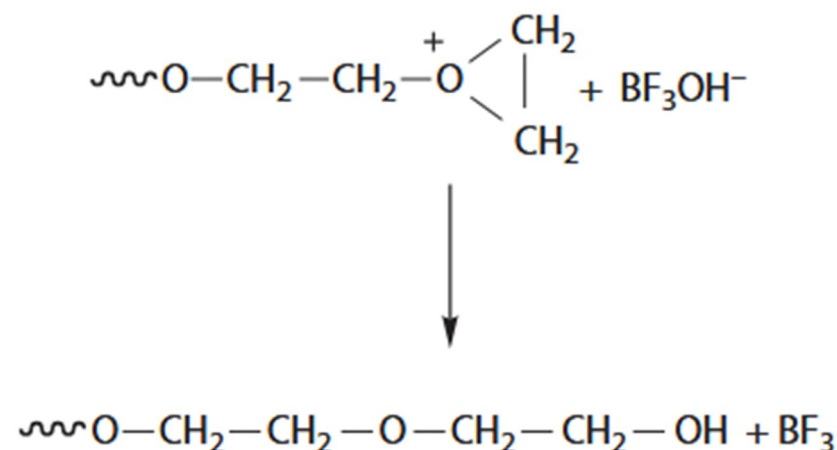
Some cROP variants are living but there are often inherent side-reactions.

At high monomer conversions, i.e., [M] low, these reactions lead to termination, rather than transfer.

Termination by back-biting and ring expansion



Termination by collapse of the counter-anion



Ring-Opening Polymerizations

Cationic Ring-Opening Polymerization (cROP)

Examples

Polytetrahydrofuran



Ring-Opening Polymerizations

Cationic Ring-Opening Polymerization (cROP)

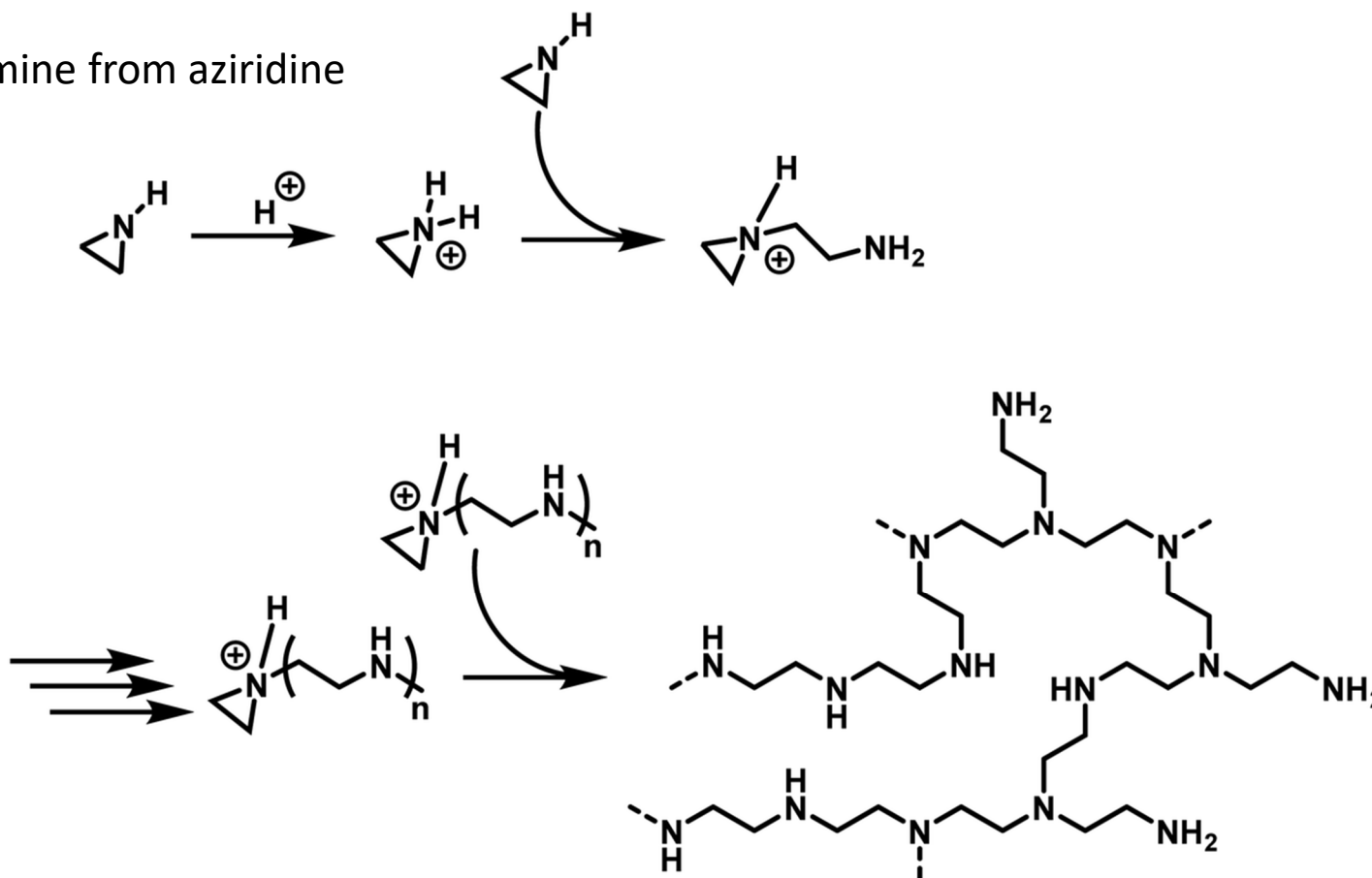
Examples

Polyoxymethylene



Examples

Polyethyleneimine from aziridine



Linear PEI via N-substituted aziridines and 2-alkyl-2-oxazolines...

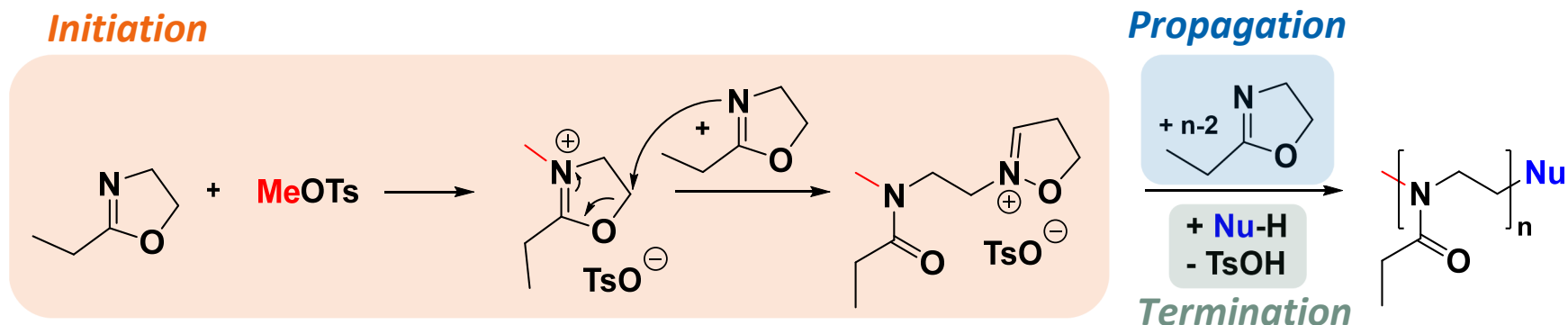
Polym. Chem. **2019**, *10*, 3257

Ring-Opening Polymerizations

Cationic Ring-Opening Polymerization (cROP)

Examples

Poly(2-alkyl-2-oxaz(ol)ine)s

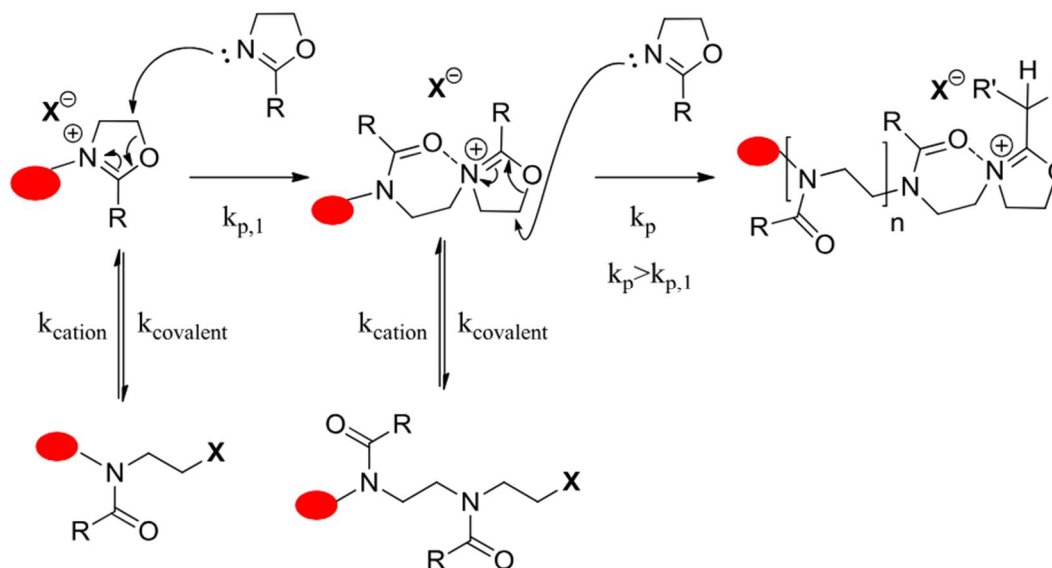


Ring-Opening Polymerizations

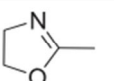
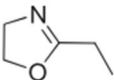
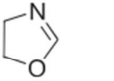
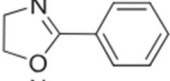
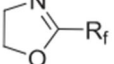
Cationic Ring-Opening Polymerization (cROP)

Examples

Poly(2-alkyl-2-oxaz(ol)ine)s



Overview of the dominant propagating species during cROP of 2-oxazolines, which depends on nucleophilicity of the monomer and nucleofugicity of the counter ion (initiator). The nucleophilicity of the monomer decreases downwards and the nucleofugicity of the counter ion increases from left to right; R_f represents a perfluorated alkyl side chain.

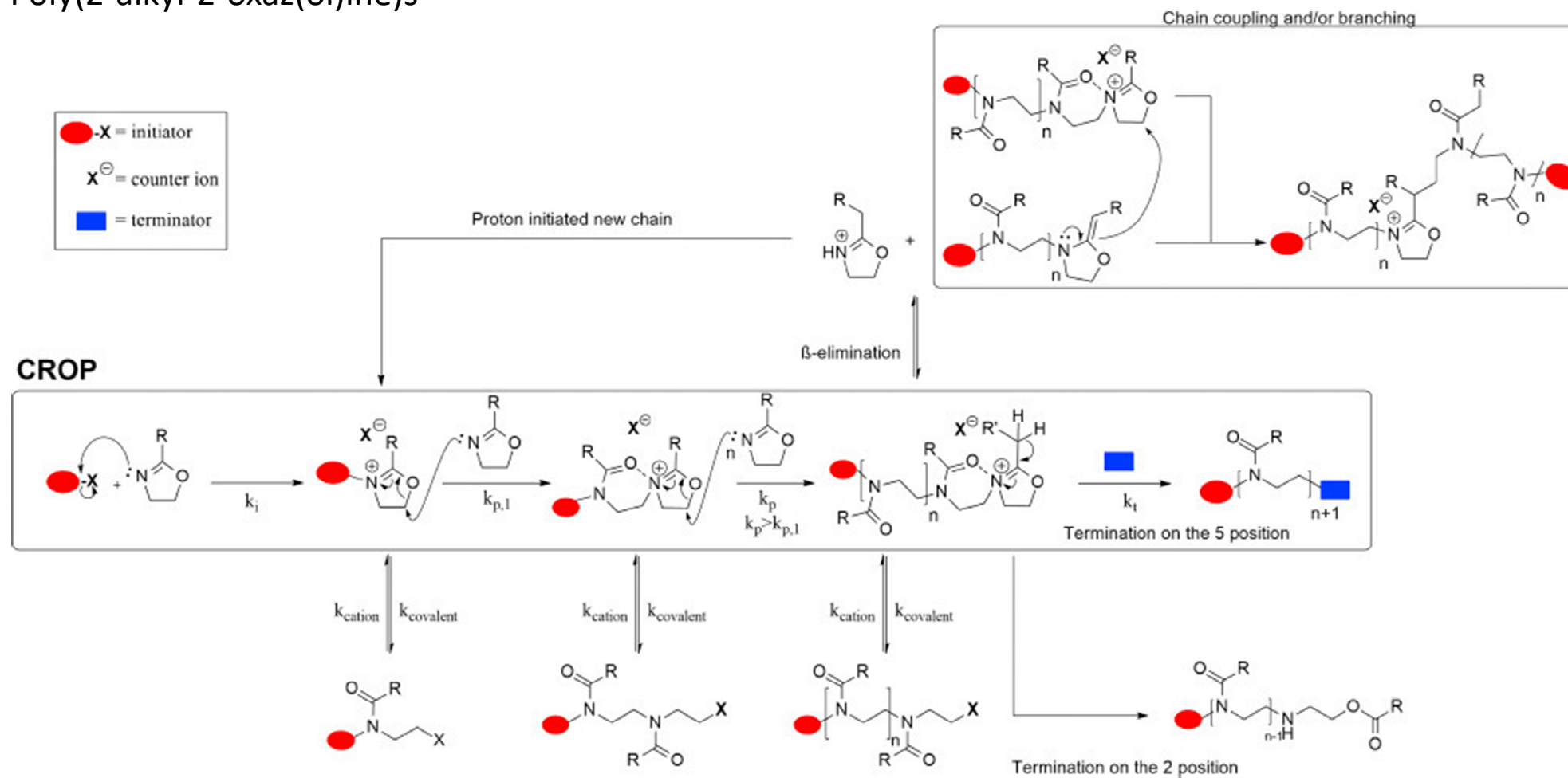
	Cl^-	Br^-	I^-	OTs^-	OTf^-
	Covalent [120]	Ionic [71,120],	Ionic [71,120]	Ionic [120,121]	Ionic [71]
	Covalent [119]	Covalent and ionic [118,119,122]	Covalent and ionic [119,123],	Ionic [119,124]	Ionic [71]
	Covalent [122]	Covalent [122]	Covalent and ionic [122]	Ionic [122]	Ionic [122]
	- [36,71]	Covalent [119]	Covalent and ionic [27,36]	Ionic [36,125]	Ionic [71]
	-	-	Covalent [27]	Covalent [126,127]	Ionic [126,127]

Ring-Opening Polymerizations

Cationic Ring-Opening Polymerization (cROP)

Examples

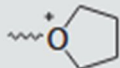
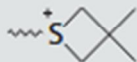
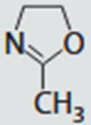
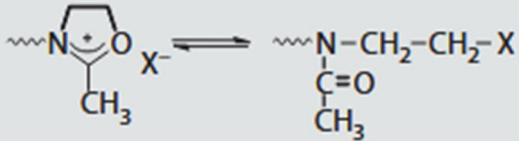
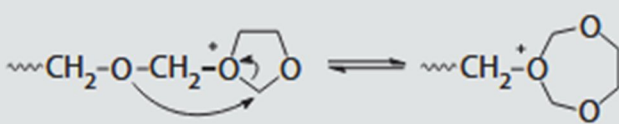
Poly(2-alkyl-2-oxaz(ol)ine)s



Ring-Opening Polymerizations

Cationic Ring-Opening Polymerization (cROP)

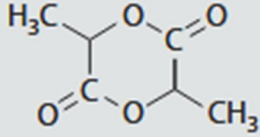
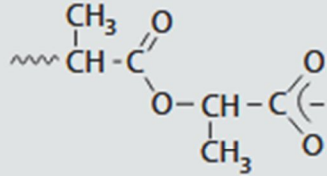
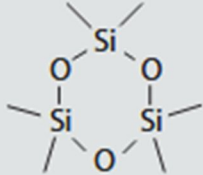
Classic structures of monomers and active chain ends in cROP

Monomer	Structure of the growing chain end
	
	
	
	

Ring-Opening Polymerizations

Anionic Ring-Opening Polymerization (aROP)

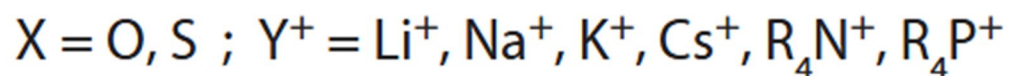
Classic structures of monomers and active chain ends in aROP

Monomer	Structure of the growing chain end
	$\sim\text{CH}_2-\text{CH}_2-\text{O}^-$
	$\sim\text{CH}_2-\text{CH}_2-\text{S}^-$
	$\sim\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{S}^-$
	
	$\sim\text{Si}(\text{R})_2-\text{O}-\text{Si}(\text{R})_2-\text{O}-\text{Si}(\text{R})_2-\text{O}^-$

Ring-Opening Polymerizations

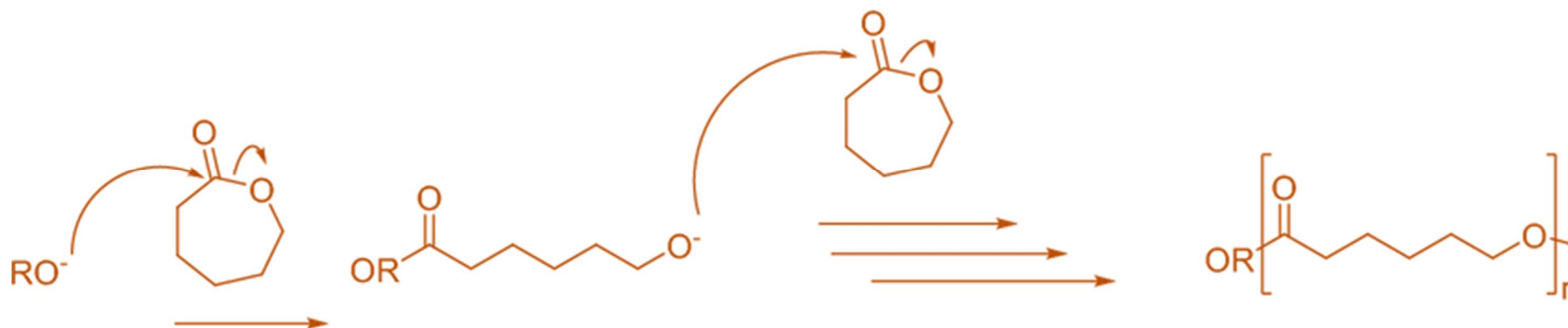
Anionic Ring-Opening Polymerization (aROP)

aROP = nucleophilic reaction of an anionic chain end with a heterocyclic monomer



Examples

Poly(ϵ -caprolactone)

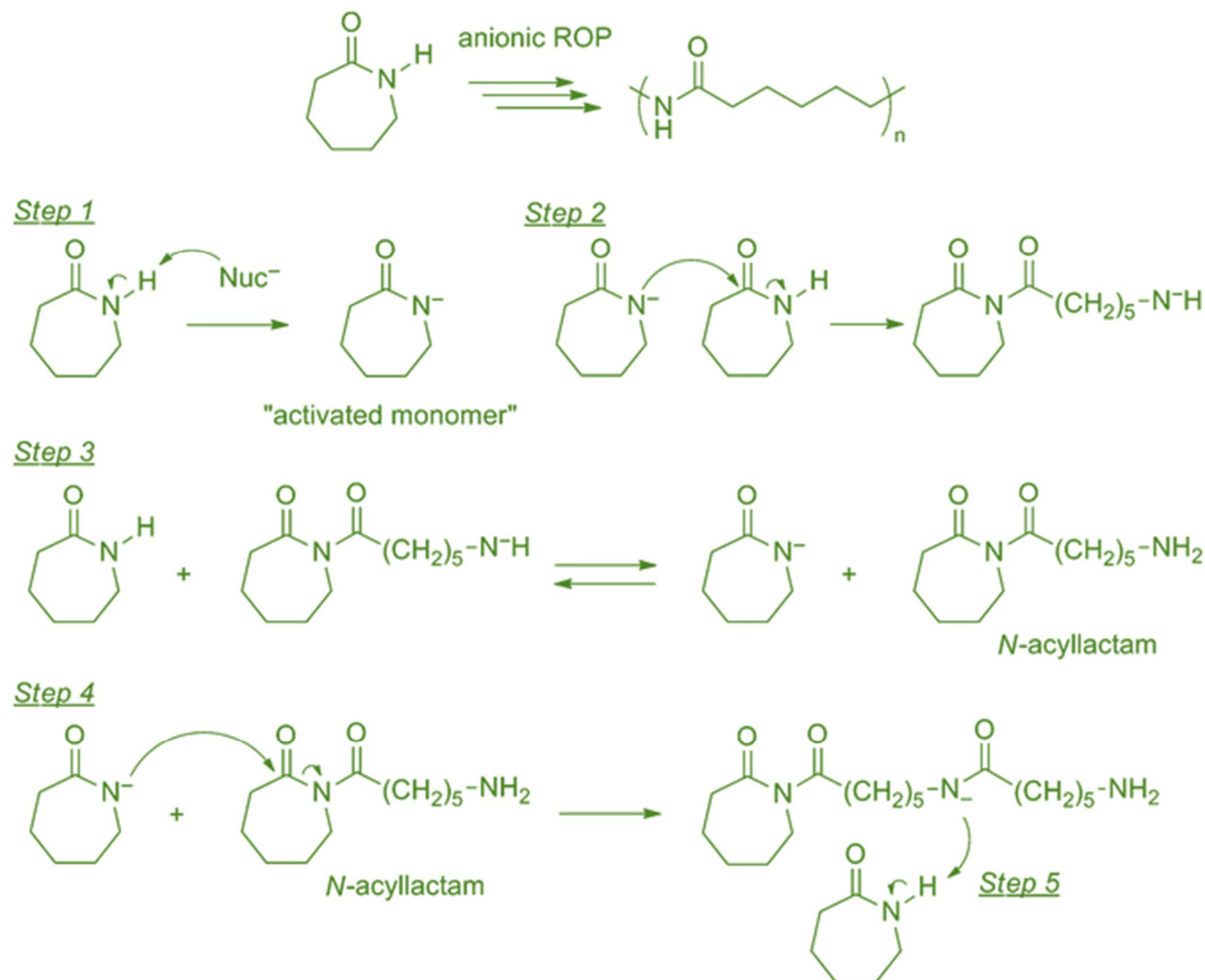


Ring-Opening Polymerizations

Anionic Ring-Opening Polymerization (aROP)

Examples

Polycaprolactam

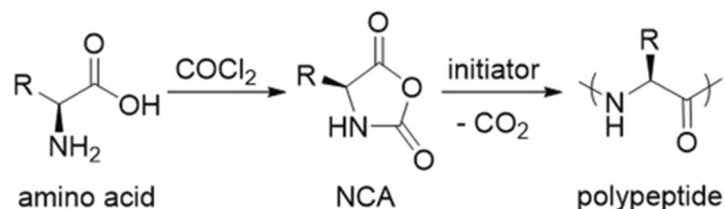


Ring-Opening Polymerizations

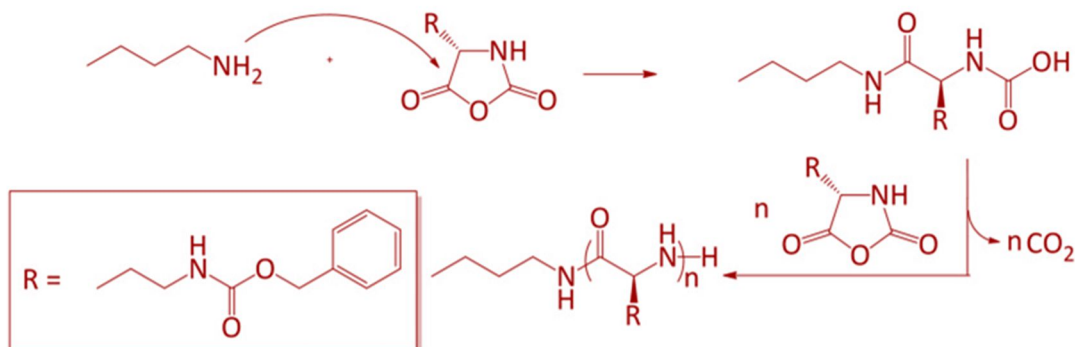
Anionic Ring-Opening Polymerization (aROP)

Examples

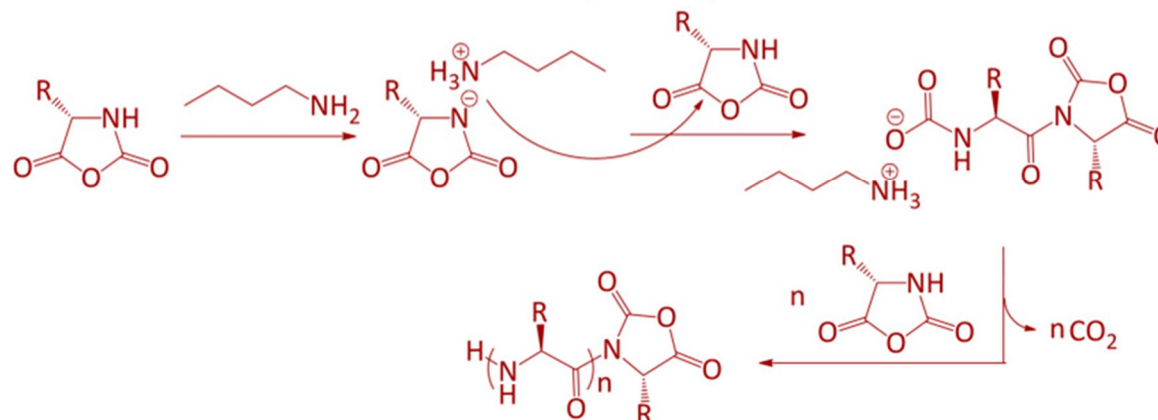
Polyamides via *N*-Carboxyanhydrides



Normal Amine Mechanism (NAM)



Activated Monomer Mechanism (AMM)



Ring-Opening Polymerizations

Cationic Ring-Opening Polymerization (cROP)

Examples

Polyethylene oxide (polyethylene glycol)



Ring-Opening Polymerizations

Cationic Ring-Opening Polymerization (cROP)

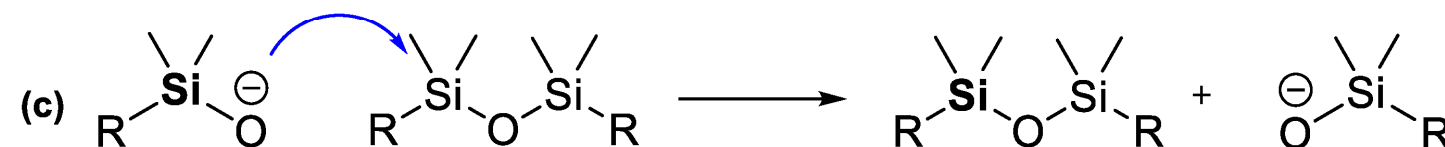
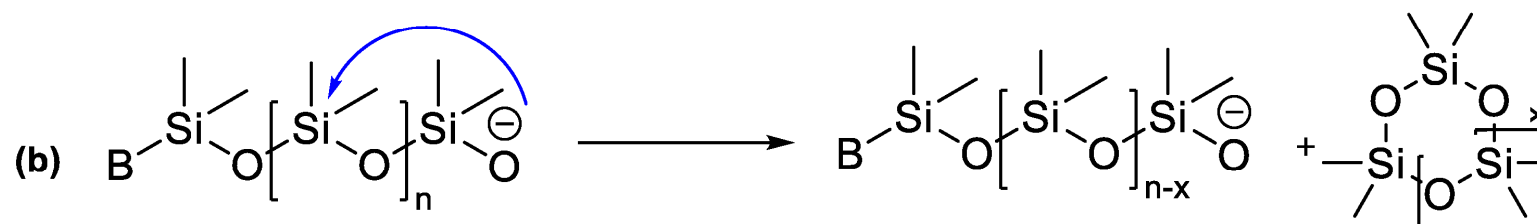
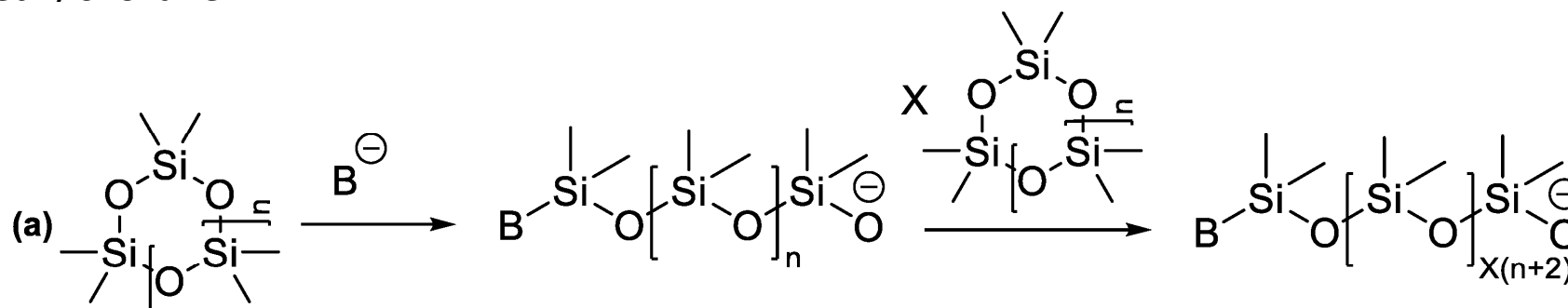
Examples

Polypropylene oxide (polypropylene glycol)



Examples

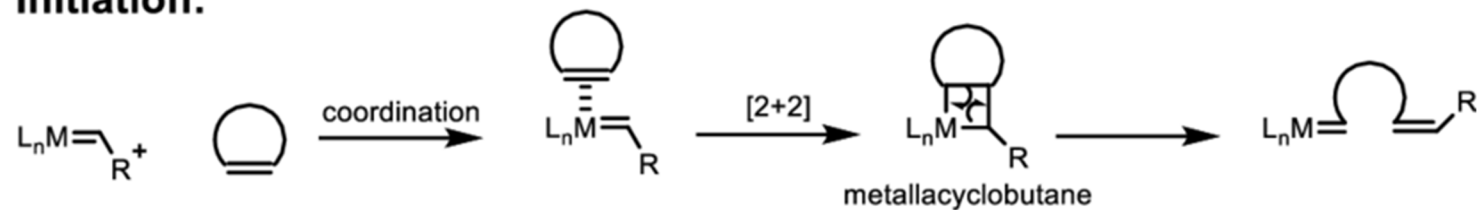
Polydimethylsiloxane



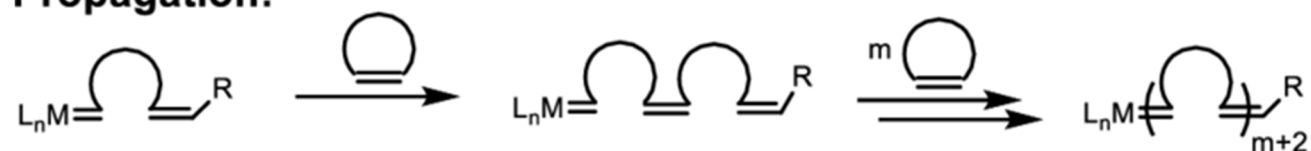
Ring-Opening Polymerizations

Ring-Opening Metathesis Polymerization (ROMP)

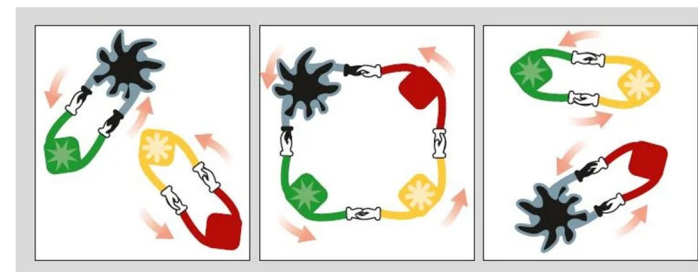
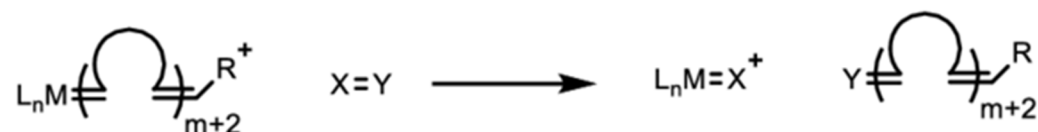
Initiation:



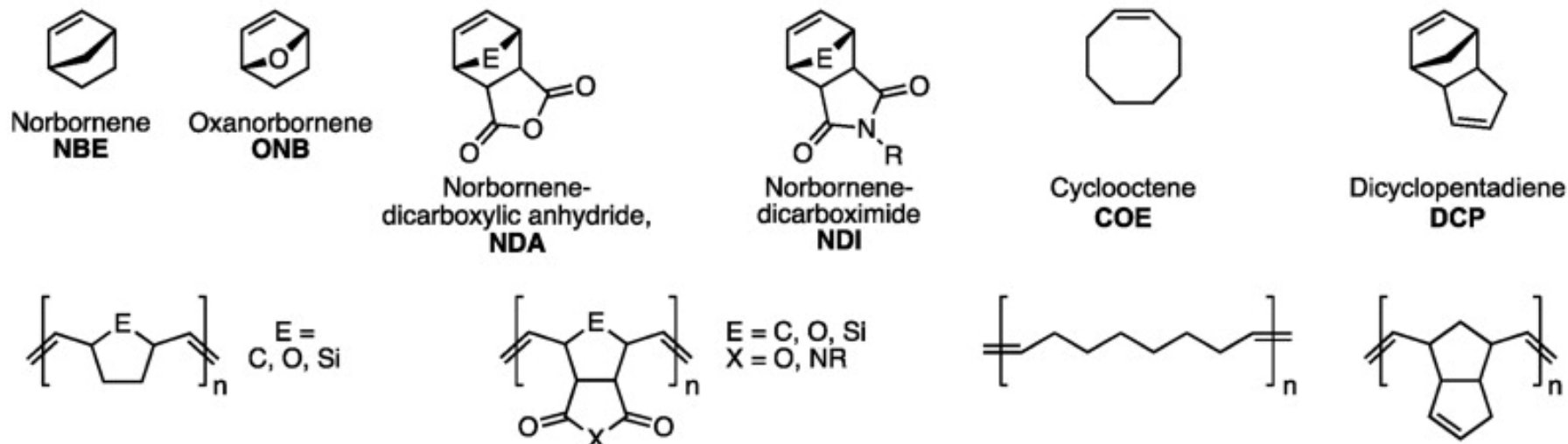
Propagation:



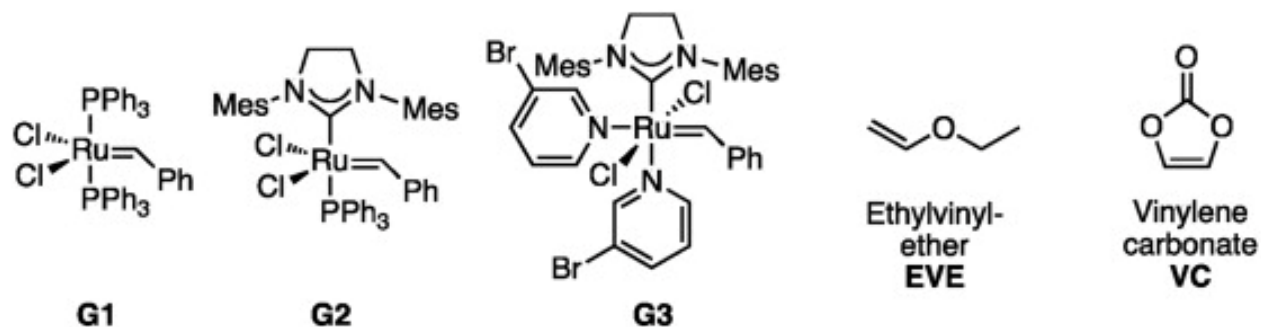
Termination:



(A) Common ROMP-polymerizable olefins with polymerized structures



(B) Typical Initiators, terminating agents



Ring-Opening Polymerizations

Ring-Opening Metathesis Polymerization (ROMP)

- Cross Metathesis
- Acyclic Diene Metathesis (ADMET)
- Enyne Metathesis
- Asymmetric Metathesis
- **Ring-opening Metathesis Polymerization**
- **Ring-closing Metathesis**

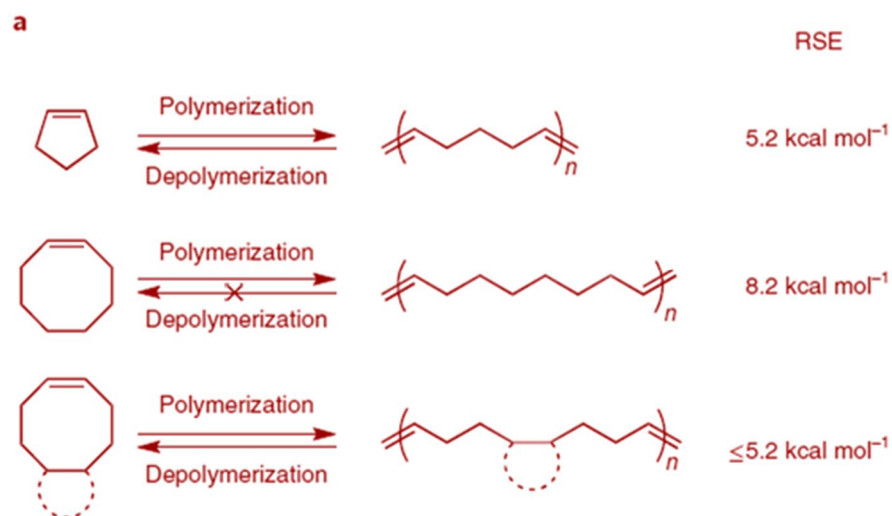


Example

Depolymerizable polycyclooctene

Fig. 1 | Identifying the appropriate ring that lowers the RSE for cyclooctene to enable depolymerization of the corresponding polymer.

a, Polycyclopentene is depolymerizable, whereas polycyclooctene is not. The difference is caused by a higher RSE in cyclooctene. An appropriate ring (shown as a dashed line) lowers the RSE of cyclooctene to a level that is lower than or comparable to that of cyclopentene, enabling depolymerization of the polymer. **b**, The calculated RSEs for cyclooctene with 3-, 4-, 5- and 6-membered rings both *cis*- and *trans*-fused at the 5 and 6 positions. **c**, The generalized ring-closing metathesis reaction used to calculate the RSE of a cyclic olefin: the RSE for the cyclic olefin is essentially the enthalpy change for the ring-closing metathesis that affords the cyclic olefin.



Consideration of ring-strain energy (RSE)

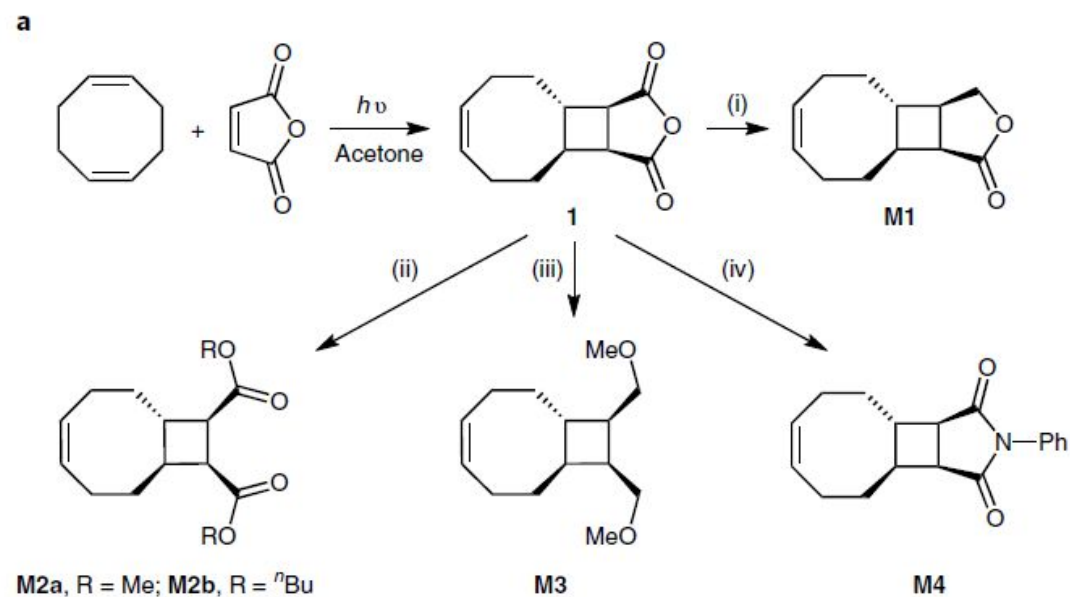


b

	3-membered	4-membered	5-membered	6-membered
<i>Cis</i>	 9.8 kcal mol ⁻¹	 10.3 kcal mol ⁻¹	 7.6 kcal mol ⁻¹	 7.2 kcal mol ⁻¹
<i>Trans</i>	 10.3 kcal mol ⁻¹	 4.9 kcal mol ⁻¹	 5.3 kcal mol ⁻¹	 7.5 kcal mol ⁻¹

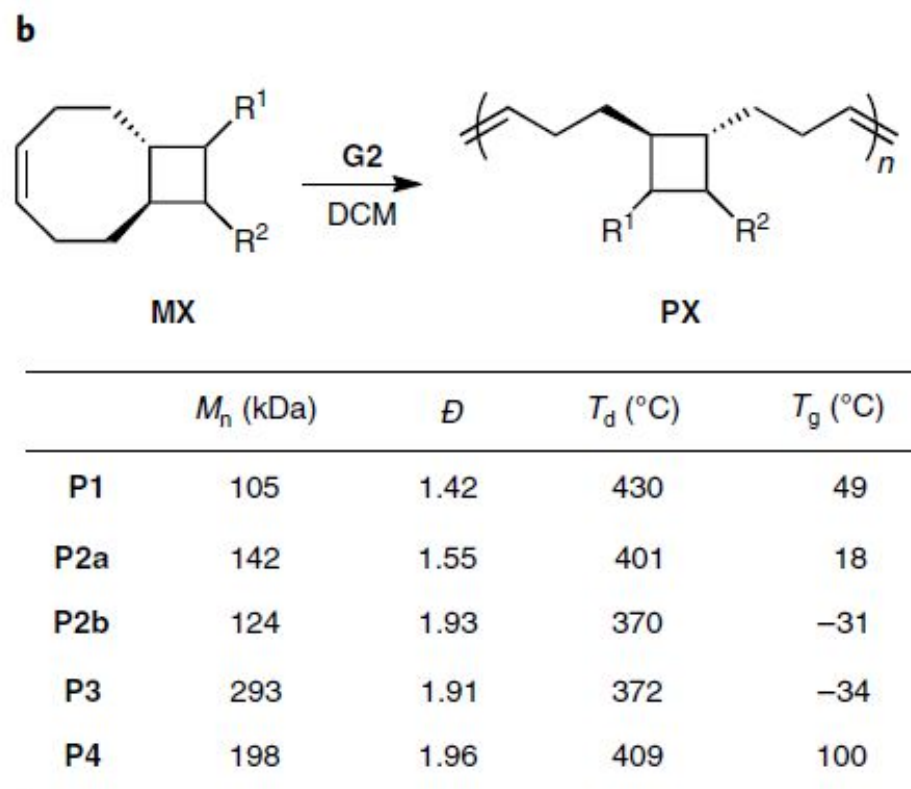
Example

Depolymerizable polycyclooctene



Example

Depolymerizable polycyclooctene



b, Synthetic scheme, molecular weight information and thermal characterization data of the tCBCO polymers. THF, tetrahydrofuran; EDC, 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride; DMAP, 4-dimethylaminopyridine; DCM, dichloromethane.

Example

Depolymerizable polycyclooctene

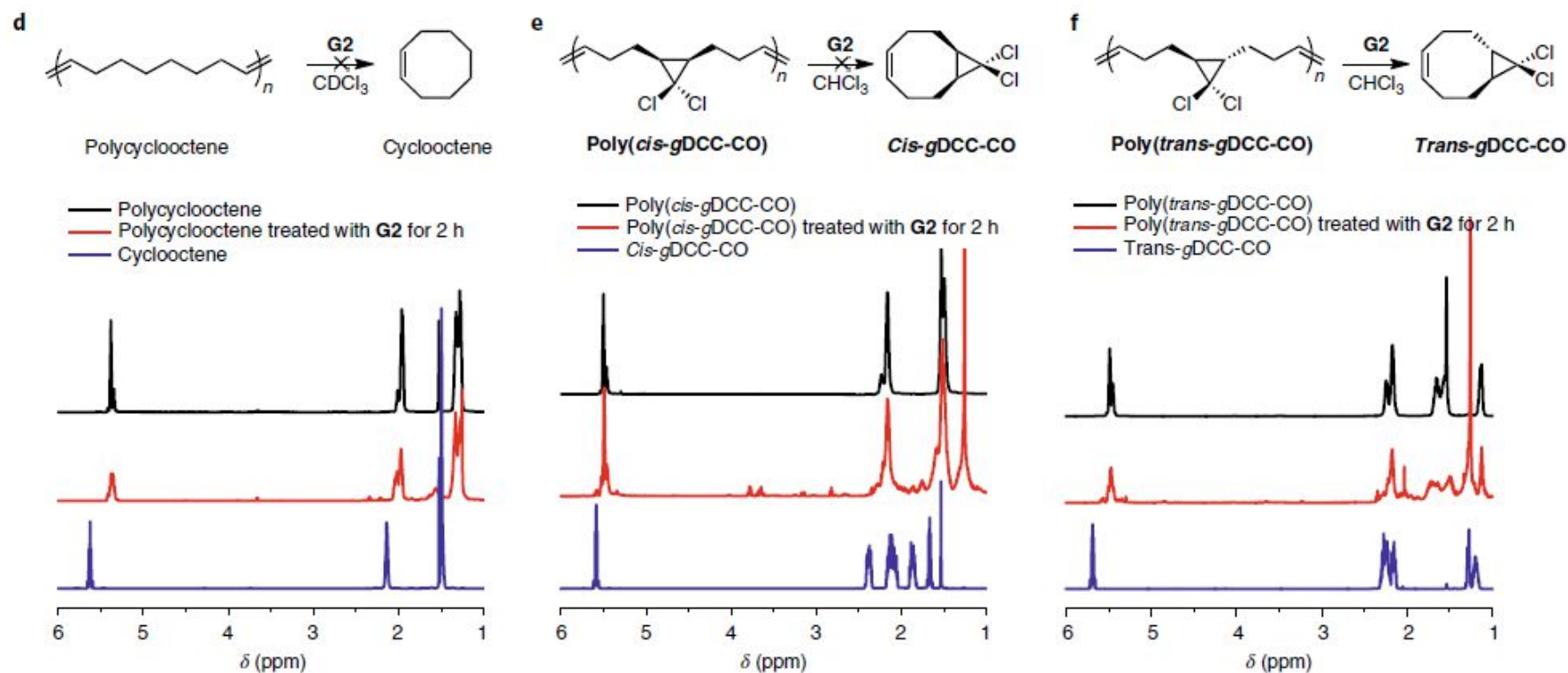


Fig. 3 | Depolymerization studies of tCBCO polymers. a–f, The ^1H NMR spectra for P1 (a), P2a (b), P3 (c), polycyclooctene (d), poly(*cis*-gDCC-CO) (e) and poly(*trans*-gDCC-CO) (f) before (in black) and after (in red) 2 h of heating the polymer solution (solvent, chloroform or deuterated chloroform; [olefin] = 25 mM) at 50 °C in the presence of Grubbs's second-generation catalyst (G2). The ^1H NMR spectra of the corresponding monomers are shown (in blue) as references. The red spectra (polymer treated with G2) and blue spectra (monomer) are nearly identical for tCBCO polymers, indicating complete depolymerization (a–c); however, the spectra are distinct for polycyclooctene, poly(*cis*-gDCC-CO) and poly(*trans*-gDCC-CO), suggesting no depolymerization occurred (d–f).

Example

Depolymerizable polycyclooctene

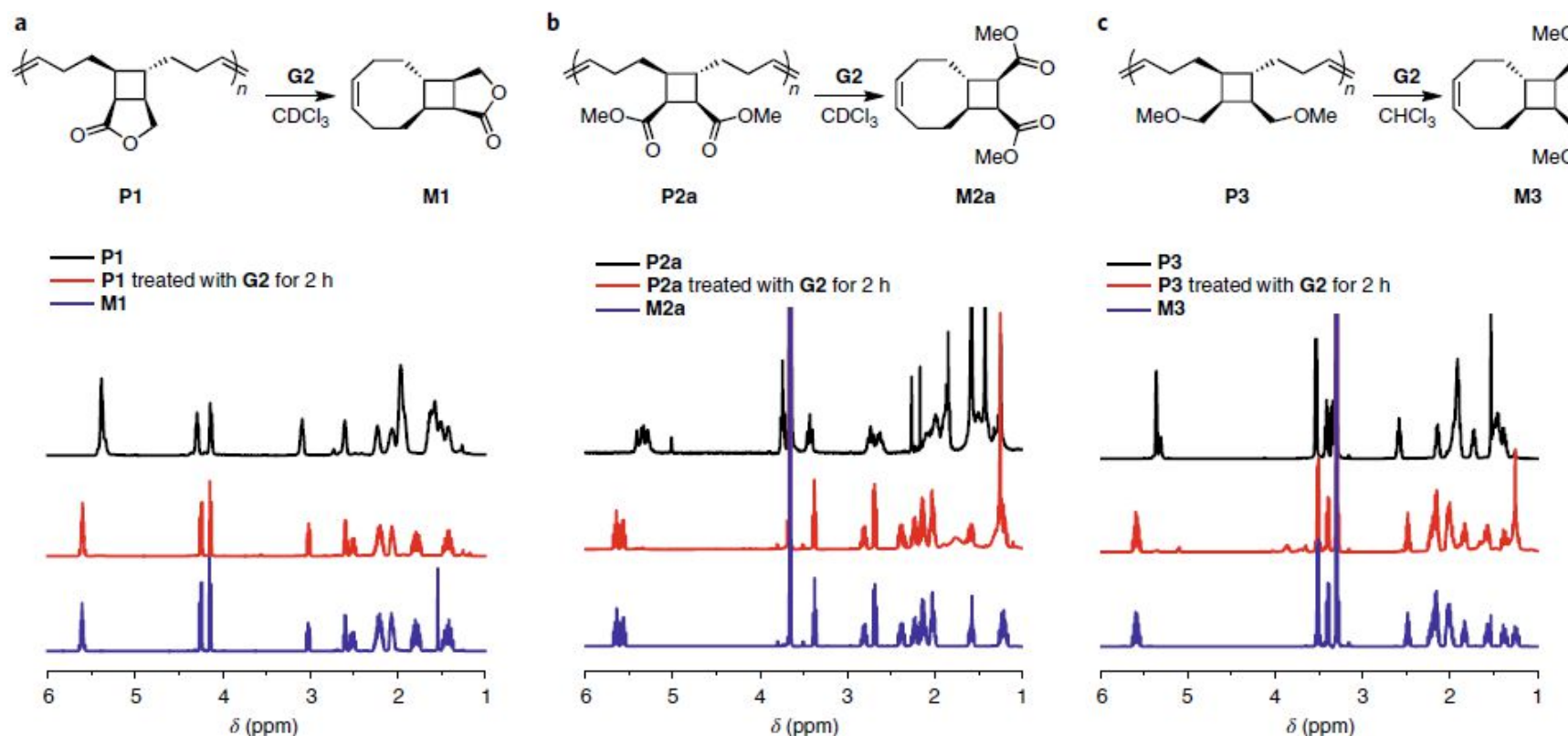


Fig. 3 | Depolymerization studies of tCBCO polymers. a–f, The ^1H NMR spectra for **P1** (a), **P2a** (b), **P3** (c), polycyclooctene (d), poly(*cis*-gDCC-CO) (e) and poly(*trans*-gDCC-CO) (f) before (in black) and after (in red) 2 h of heating the polymer solution (solvent, chloroform or deuterated chloroform; [olefin] = 25 mM) at 50 °C in the presence of Grubbs's second-generation catalyst (**G2**). The ^1H NMR spectra of the corresponding monomers are shown (in blue) as references. The red spectra (polymer treated with **G2**) and blue spectra (monomer) are nearly identical for tCBCO polymers, indicating complete depolymerization (a–c); however, the spectra are distinct for polycyclooctene, poly(*cis*-gDCC-CO) and poly(*trans*-gDCC-CO), suggesting no depolymerization occurred (d–f).

Nat. Chem. 2021, 13, 743

Advantages

- Often – but really not always – polymerizations at low temperatures
- Except for rROP, access to some functional groups that are incompatible with radicals (nitroxides, nitro), but of course just as any non-radical polymerizations
- Synthesis of polymers with functional backbones → Access to polymers with a broad spectrum of degradation or depolymerization modes

Drawbacks

- Significantly less commercially available monomers than for vinyl polymerizations
- Overall lower compatibility with functional groups